WEEK 12 PROJECT REPORT

Automobile Analytics Dashboard

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1. Introduction

This project shows how we built an interactive Automobile Analytics Dashboard in Microsoft Power BI. Using the clean data model and DAX measures created in Week 11, the main goal for Week 12 is to visually explore vehicle performance, pricing, fuel efficiency, and technical specifications.

We created KPI cards for overall averages, charts to compare engine power against price and fuel mileage, a comparison matrix to group cars by body style and drive wheels, and a custom hover tooltip page to show full vehicle details.

2. Objectives

• Display key average figures (Avg Price, Avg Horsepower, Avg Engine Size, and Avg Highway MPG) using KPI Cards.

• Create a Scatter Plot mapping horsepower vs. price with engine-size bubble scaling and color coding by body-style.

• Build a Comparison Matrix grouping vehicles by body-style and drive-wheels alongside our average DAX measures.

• Create Column Charts to compare city-mpg vs. highway-mpg and analyze vehicle manufacturers (make).

• Build a custom Tooltip (Vehicle Tooltip) displaying full technical details when hovering over data points.

3. Tools Used

• Microsoft Power BI Desktop: The main software used to design the dashboard, build charts, and format visuals.

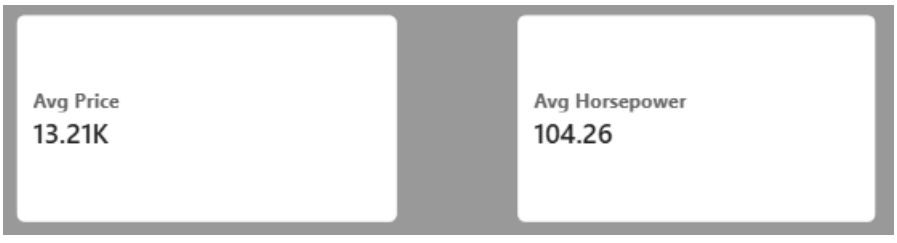
• DAX Measures (_Measures): The custom average formulas built in Week 11 (Avg Price, Avg Horsepower, Avg Engine Size, Avg Highway MPG).

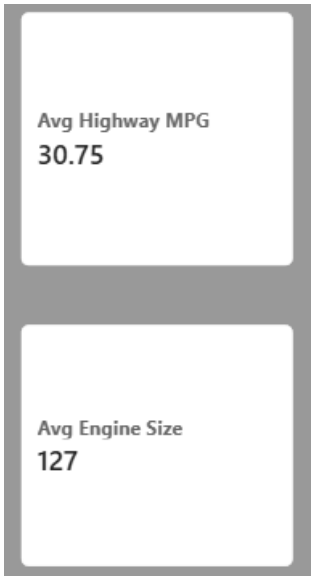
• Report Page Tooltips: The Power BI feature used to create a hidden pop-up technical card for individual vehicles.

4. Step-by-Step Work

Step 1 – Creating KPI Cards

• Add four Card visuals across the dashboard using the measures from our _Measures table to display the overall averages for Price, Horsepower, Engine Size, and Highway MPG.

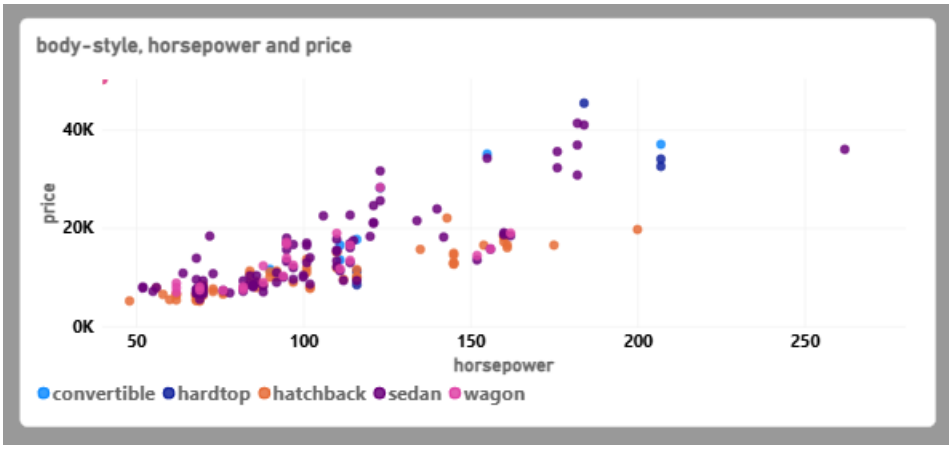




Step 2 – Building the Scatter Plot

• Add a Scatter chart mapping horsepower on the X-axis and price on the Y-axis (both set to Don't summarize so every car appears individually).

• We added engine-size to control bubble size and body-style to color-code the vehicles.

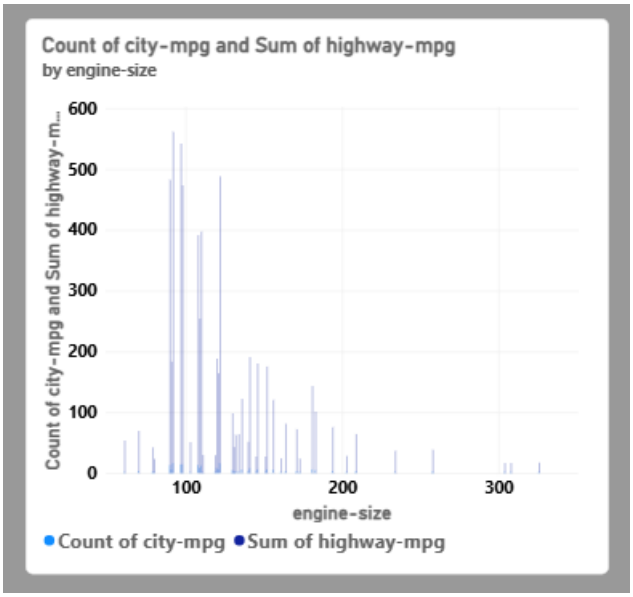


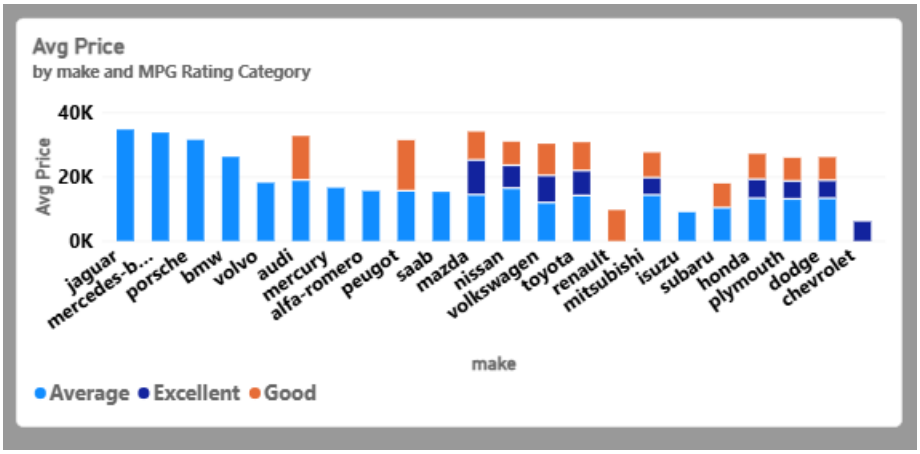
Step 3 – Creating the Comparison Matrix and Column Charts

• Add Matrix visual grouping body-style and drive-wheels in the rows, with our four average DAX measures in the values.

body-style	Avg Price	Avg Horsepower	Avg Highway MPG	Avg Engine Size
convertible	21,890.50	131.67	26.00	157
hardtop	22,208.50	142.25	27.25	176
hatchback	9,957.44	101.33	32.17	114
sedan	14,459.76	103.10	30.83	131
wagon	12,371.96	97.75	28.72	124
Total	13,207.13	104.26	30.75	127

• We created Clustered Column Charts to compare City MPG by Highway MPG and evaluate vehicle manufacturers (make) across segments.



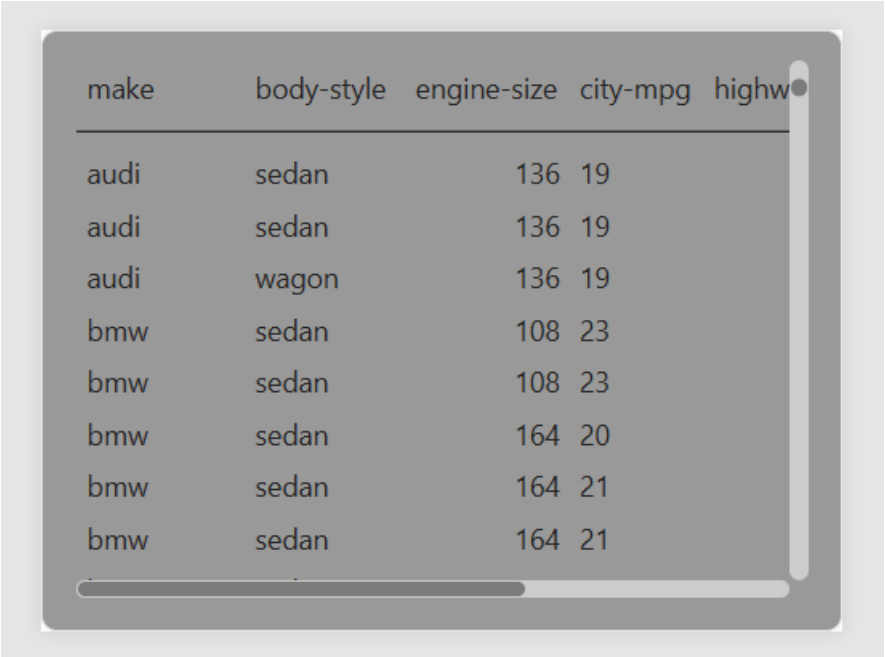


Step 4 – Building the Custom Vehicle Tooltip

• Create a second page named Vehicle Tooltip, enabled Allow use as tooltip, and set the canvas size to Tooltip.

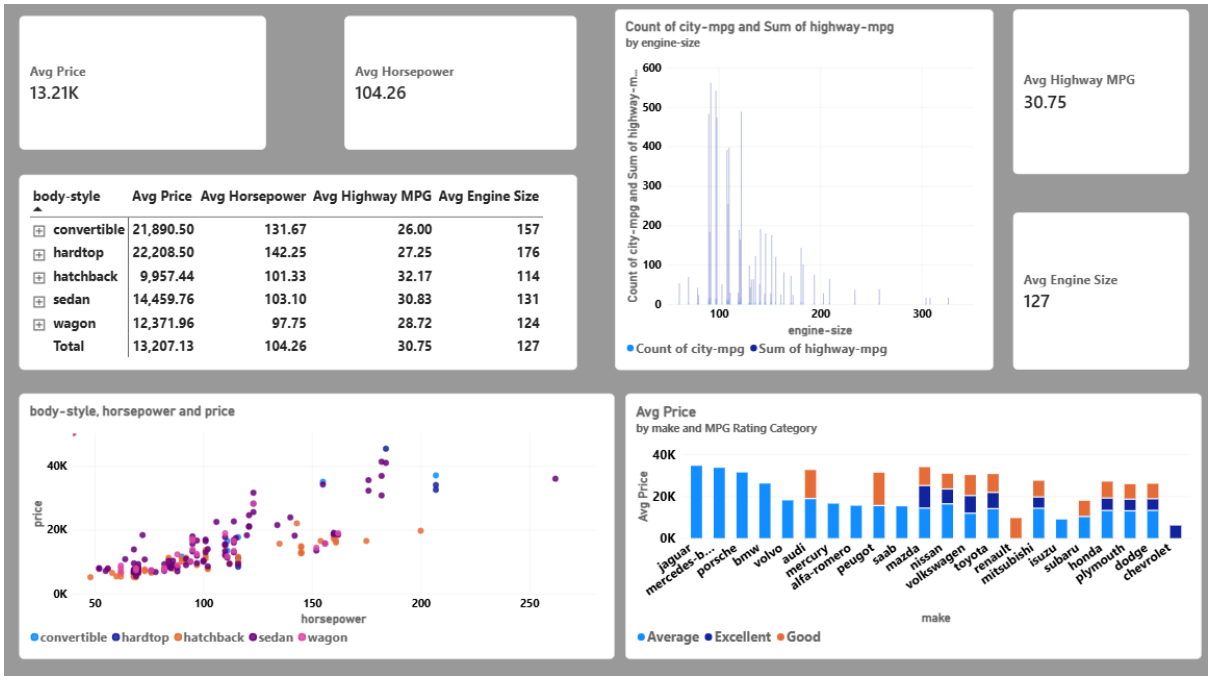
• Add a Table visual showing exact vehicle details (make, body-style, engine-size, city-mpg, highway-mpg, horsepower, and price).

• We linked this tooltip page to our Scatter Plot.



make	body-style	engine-size	city-mpg	highw
audi	sedan	136	19	
audi	sedan	136	19	
audi	wagon	136	19	
bmw	sedan	108	23	
bmw	sedan	108	23	
bmw	sedan	164	20	
bmw	sedan	164	21	
bmw	sedan	164	21	

Step 5 – Final Dashboard Layout



5. Conclusion

In Week 12, we successfully turned our clean automobile data model into an interactive Power BI dashboard. Using KPI cards, scatter plots, comparison matrices, and custom hover tooltips, users can easily analyze how engine size and horsepower impact vehicle price and fuel efficiency across different car styles and manufacturers.